Tecnológico de Monterrey

MODELING OF ENGINEERING WITH COMPUTATIONAL MATHEMATICS

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INDEX

1. Introduction 3
2. Credits 3
3. Context and Problem 3
4. General objectives 3
5. Survey Design 3
6. Data Analysis 4
6.1 Quantitative analysis of the survey (at least two questions per person) 4
7. Product proposal 4
7.1 Sketch of the proposed solution 4
8. Model in Matlab 4
8.1 Model code 4
8.2 Matlab model vs. original proposal 4
9. Cost analysis 4
10. Proposal for technological innovation (at least one proposal per person) 4
11. Project Conclusions and Insights (individual) 5
12. Functional Prototype 5
13. Project Management Document 5
14. References 5
15. Appendices 5

1. Introduction

In recent years, pets—especially dogs—have become an integral part of modern families. Many households now consider dogs as family members rather than just pets. However, lifestyle changes, urbanization, and work responsibilities have significantly altered the daily routines of both owners and their pets.

Most dogs today live in urban environments such as small houses or apartments, where space for physical activity is limited. Additionally, research published in *Scientific Reports* indicates that most dog owners work approximately 40 hours per week (Nature, 2025). This means that many dogs spend long periods alone at home without adequate physical or mental stimulation.

Lack of stimulation is directly linked to behavioral problems. According to Poian (2026), approximately 67% of behavioral issues in dogs are symptoms of chronic under-stimulation rather than disobedience. When dogs do not receive enough mental and physical engagement, they may develop anxiety, destructive behaviors, excessive barking, and even health problems related to stress or inactivity. Furthermore, mental stimulation has been shown to improve cognitive function, reduce boredom, and promote overall well-being in dogs (Dogster, n.d.).

The specific situation our team aims to solve is the lack of daily stimulation for dogs whose owners are busy working or studying. Many owners want to provide better enrichment for their pets but do not always have the time during the day to actively engage in play.

To address this problem, our team proposes Bark & Play, an interactive and eco-friendly dog toy designed to stimulate movement and reward engagement. The toy incorporates a built-in treat-dispensing system that encourages dogs to move, chase, and interact with the product. By combining physical activity with positive reinforcement, Bark & Play promotes healthy behavior while keeping dogs entertained during long hours alone.

Our solution strategy includes:

- Conducting market research to validate demand.
- Designing and testing a durable, bite-resistant prototype.
- Using safe, BPA-free, and eco-friendly materials.
- Developing a digital marketing strategy to reach urban pet owners.
- Launching the product through online sales channels.

Through this approach, Bark & Play seeks to improve dogs' quality of life while providing owners with a practical and innovative solution to under-stimulation.

2. CREDITS

Bark & Play Team

- **Andrés Sarmiento Felix**
    - Market research
    - Competitor analysis
    - Value proposition development
- **Massimo Brambilla Muela**
    - Product design
    - Prototype development
    - Material selection
- **Sebastián Huerta González**
    - Marketing strategy
    - Branding and social media management
    - Launch planning
- **Sebastián Noriega Olavarrieta**
    - Project Manager
    - Strategic planning
    - Team coordination
    - Financial analysis and business model development
- **Jolett Garza Martínez**
    - Cost analysis
    - Sales projections
    - Economic feasibility evaluation

3. Context and Problem

Dogs have become a recurrent variable in modern families. But, due to the recent economy is very common for both partners in a relationship what causes not only for the dog to spend may hours alone but owners come home tired and the average can just manage to take them for a quick 10-20 min walk what is not enough, this is what limits their fiscal and mental stimulation

Research indicates that many dog owners work around 40 hours per week (Nature, 2025), reducing the time available for their dogs. Studies say that a large percentage of behavioral problems in dogs are linked to under stimulation and not disobedience (Poian, 2026). Lack of mental stimuli can lead to anxiety, bad/destructive behaviors, and even stress related health issues. On the other hand, Mental stimulation for the dogs improves cognitive function and overall well being (Dogster, n.d.).

Our proposed solution is a toy that can produce:

Physical stimulation – Encourages movement and activity.

Mental stimulation – Promotes problem solving and brain activation.

Positive reinforcement – Rewards after treat release.

Our prototype is a ball shaped toy with an internal compartment where owners can place small treats. The dog must roll and play with the ball to release the treat. This toy successfully produces all those stimulations naturally. All this without the need of the owner to be present

4. General objectives



- The goal is creating a toy that entertains dogs during the day, while their owners are busy or at work.
- The company's goal is to sell about 5,000 toys. After one year in the market, we want our product to be sold at Petco Monterrey. The product projections indicate that the product will be used by 2%-10% of dog owners in Monterrey 2 years after we release it.

5. Survey design

Identification and explanation of the number and objective of the questions to be asked in the survey (at least 10 and no more than 25), aimed at identifying the preferences of the analysis groups. You MUST validate this design with the Module 1 Professor

https://docs.google.com/forms/d/e/1FAIpQLSczd9K_psyRrzTRLuDFDyRnEE0PnopdbI-bTLY3drgR1UZixQ/viewform?usp=publish-editor

Question	Objective of the question: What do I hope to find or measure with the question?	How will it be analyzed?
¿Tiene alguna mascota en su hogar?	See if the people questioned have any pets.	Find out what percent of people are pet owners and have an idea of our market gap.
Si respondió 'Sí' en la pregunta anterior, ¿qué tipo de mascotas tiene?	We hope to organize and separate the opinions as well as narrowing the market.	Multiple questions with multiple answers
¿Cuántos perros tiene en su hogar?	Find out if the pets have any other animals keeping them company, since it affects its energy levels.	No means that the dog probably has more energy needing a higher resistance toy.
¿Cuál es la edad promedio de sus perros?	Dogs have different capabilities depending on the age so we need to find the age in order to know how to build the prototype and of what materials.	The older the dogs are the less movement-available the toy should be meaning we would be focusing more on a toy that stays in place and doesn't require much force to be able to play with.
¿Qué tan importante considera la compra de juguetes para sus perros?	With this question we find out how willingly the owners are spending money on the product, so we can apply a baseline cost of the toy.	The higher the importance of buying a toy means that the owner has a higher chance of spending more money on the toy.
¿con qué frecuencia compra juguetes nuevos para sus perros?	We need to find a median lifetime of the current toys in the market.	If the owner says they buy toys every 4 months then we know that we need to create a toy that will last at least 4 months, this means we focus on different materials instead of wasting time searching.
¿En su opinión qué factores son más importantes para elegir un juguete para su perro? (Seleccione 3 opciones)	With this information we are able to know where we need to focus more on, to satisfy the needs of the customer.	We will use the budget in a way that we can completely satisfy the top 3 factors that the owners think are more important.
¿Cuánto estaría dispuesto a pagar por un juguete de perro de buena calidad y duradero?	We want to know a good price, that the customer would be comfortable spending.	If the average customer has a low budget to purchase dog toys, we can adjust our production costs according to their spending expectations.
¿Cuánto tiempo pasa su perro solo?	We want to know the amount of time dogs spend alone.	This way we can create a product that can help dogs stay distracted for longer time or is easy for the customer to use if they spend time with their pets.

¿Hay algún problema o molestia que encuentre con algunos de los juguetes para perros actuales?	We are hoping to find any common flaws, so that we can make sure we don't repeat them.	Every flaw that the user comments about, will be taken into consideration in the making of this product. This way we can find and avoid these flaws.

6. Data Analysis

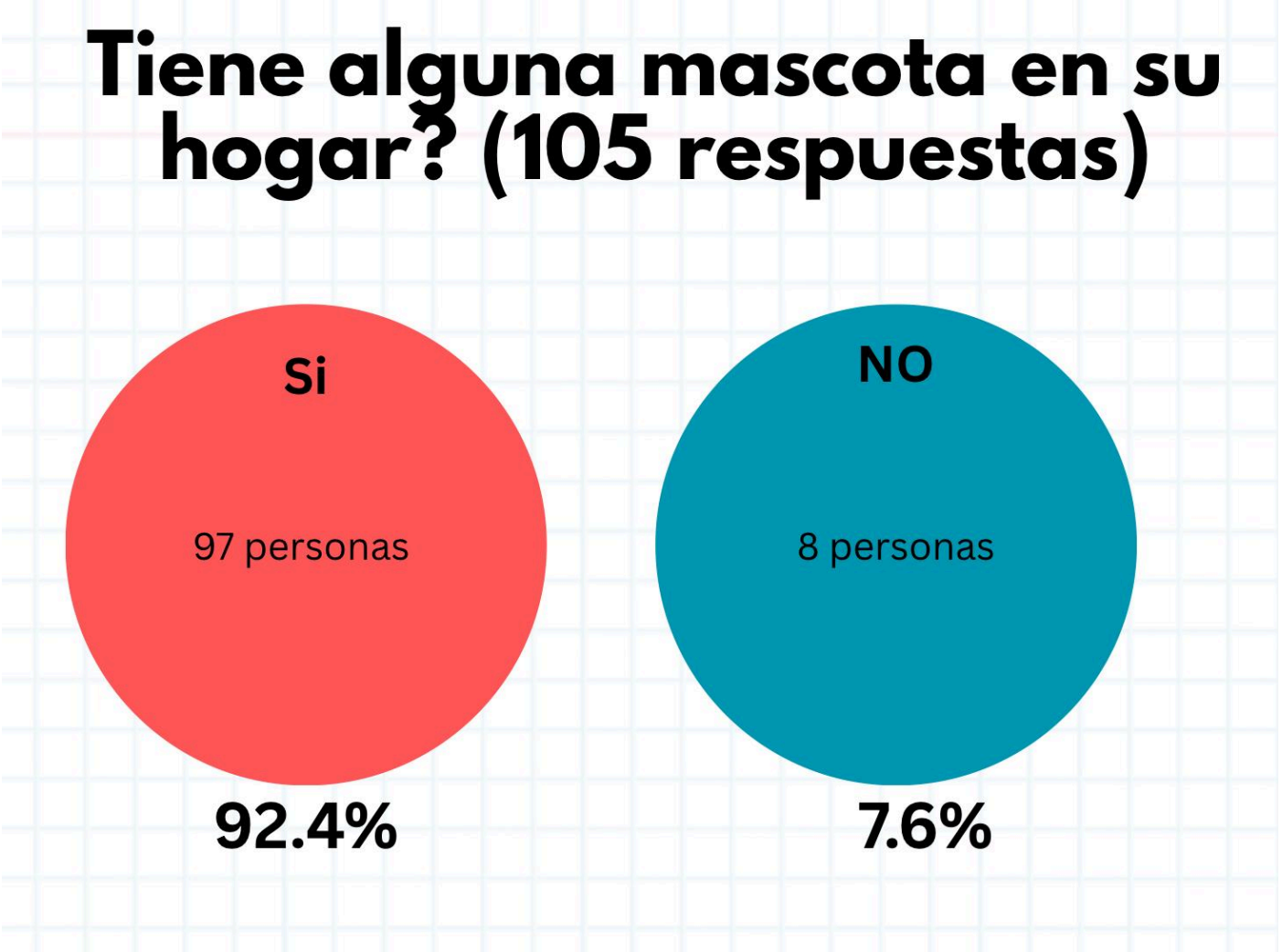
6.1 Quantitative analysis of the survey (at least two questions per person)

- Make the relevant set diagrams for each of the questions
- Make logical propositions relevant to the data obtained in the survey

Survey answers data base: Bark & Play (Respuestas)

Questions:
Andrés S:

Question 1



Question 3

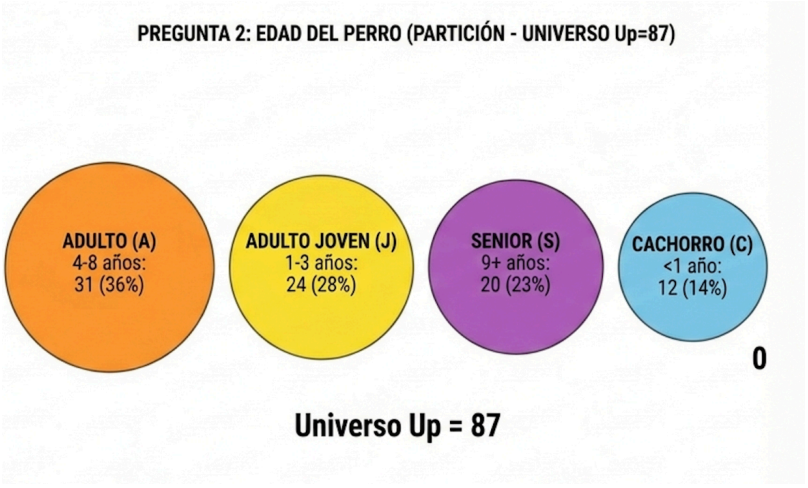
Cuantos perros tiene en su hogar?

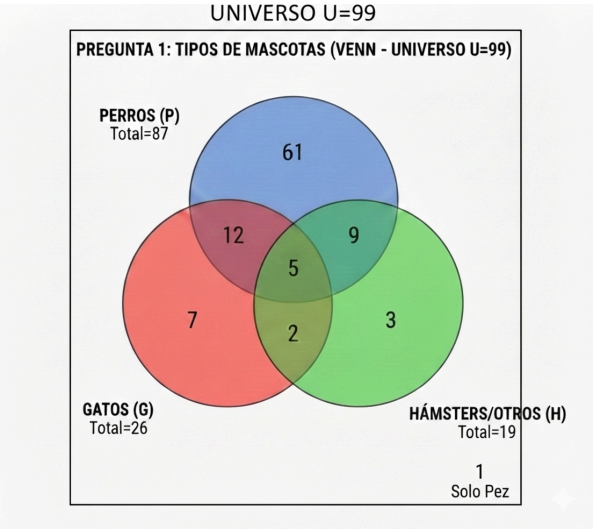


Sebastian H:

Question 2

Question 4





When the respondents were asked what type of pets they have, 99 valid responses were recorded. For the quantitative analysis, three main sets were selected for the Venn diagram: Perro(s), Gato(s), and Hámster, conejo, etc., as they allowed a clear and consistent representation of the data.

61 respondents selected having only dogs, 7 only cats, and 3 hamster, bunny, etc. Additionally, 12 reported having both dogs and cats, 9 had dogs and small mammals, and 2 had cats and small mammals. Five respondents indicated having all three types of pets. The sum of all exclusive regions equals 99, confirming the mathematical accuracy of the diagram.

Even though the survey had the categories Ave(s), Pez(es), and Otro, none of the respondents selected these options exclusively. All respondents who marked these choices also selected at least one of the three main sets that we put in

the diagrams but one did put only pez so that one wa put in the universe outside the circles. These categories were not grouped into a single “Other” set because that would mean creating another survey because grouping it would create internal overlaps because there would be double counting in that “Other” set because some respondents selected multiple options in those categories for example, 10 people marked Ave, 8 scored Pez, 5 marked Other But, 6 of Ave's also have a dog, 4 of the Pez also have a dog, 3 marked Bird and Fish, 2 marked Bird, Fish and Hamster If you do: O = 10 + 8 + 5 = 23 You are already duplicating: The 3 people who marked Bird ⋂ Fish The 2 that marked Bird ⋂ Fish ⋂ H Then the total starts to raise. So a three circle Venn diagram cannot represent those internal intersections. So the selected sets show a mathematically clear image of the data.

The second question about the age of the dog, the categories (Cachorro, Adulto joven, Adulto, Senior) are exclusive, meaning no intersections exist. So this question is represented as a categorical distribution and not with overlapping sets.

In conclusion, the sets we chosed provide a clear and consistent image of the 99 responses, with no missibg elements in the universe.

Universe: 99

EXCLUSIVE REGIONS:
Only P = 61

Only G = 7

Only H = 3

DOUBLE INTERSECTIONS:

P ⋂ G = 12

P ⋂ H = 9

G ⋂ H = 2

TRIPLE INTERSECTION

P ⋂ G ⋂ H = 5

Out of any cicle = 1

Question 2:

Universe = 87

No interceccions:

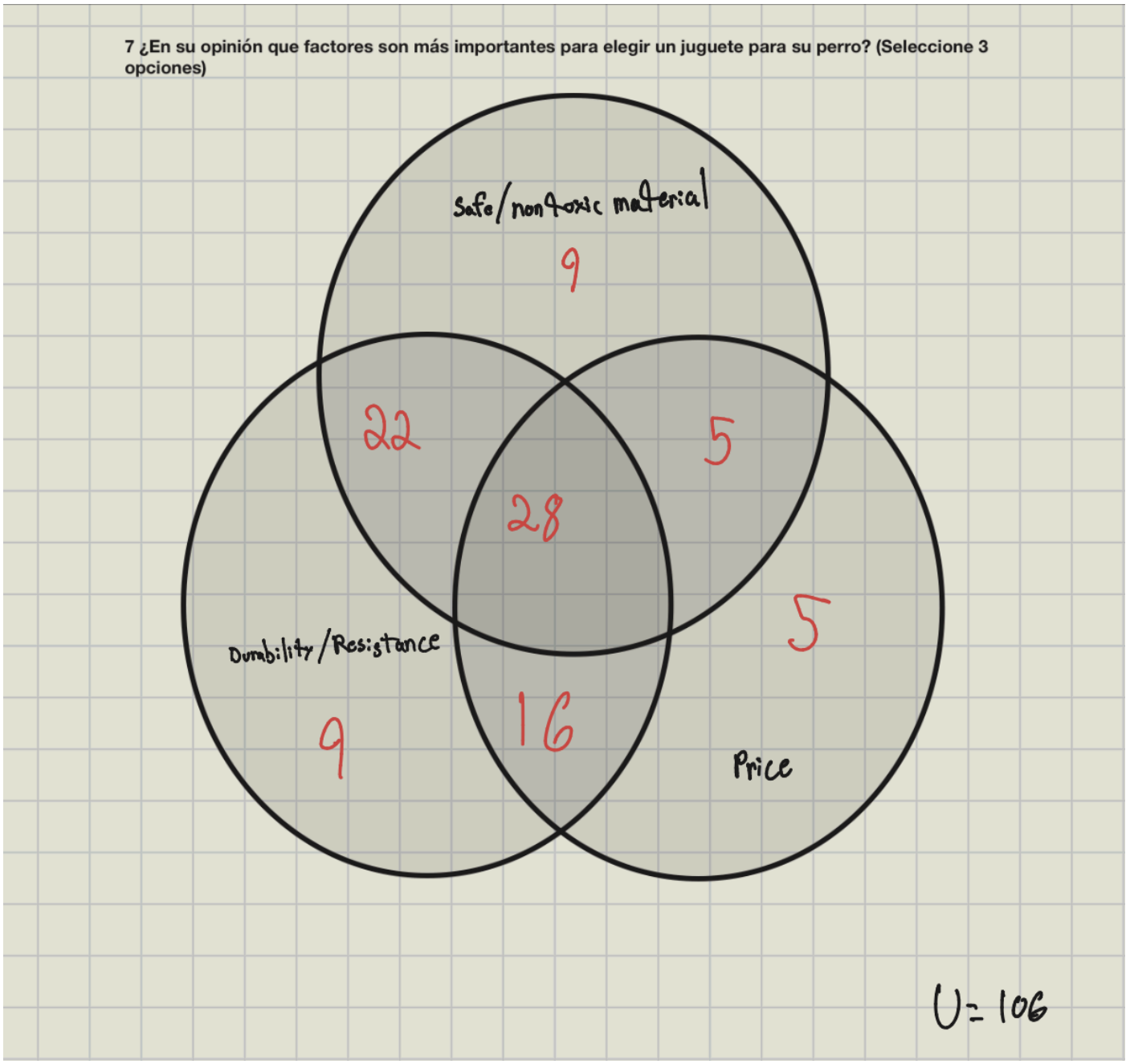
Cachorro (menos de 1 año) = 18

Adulto joven (1–3 años) = 29

Adulto (4–8 años) = 25

Senior (9 años o más) = 15

Sebastian N:



Question 7 – Quantitative Analysis

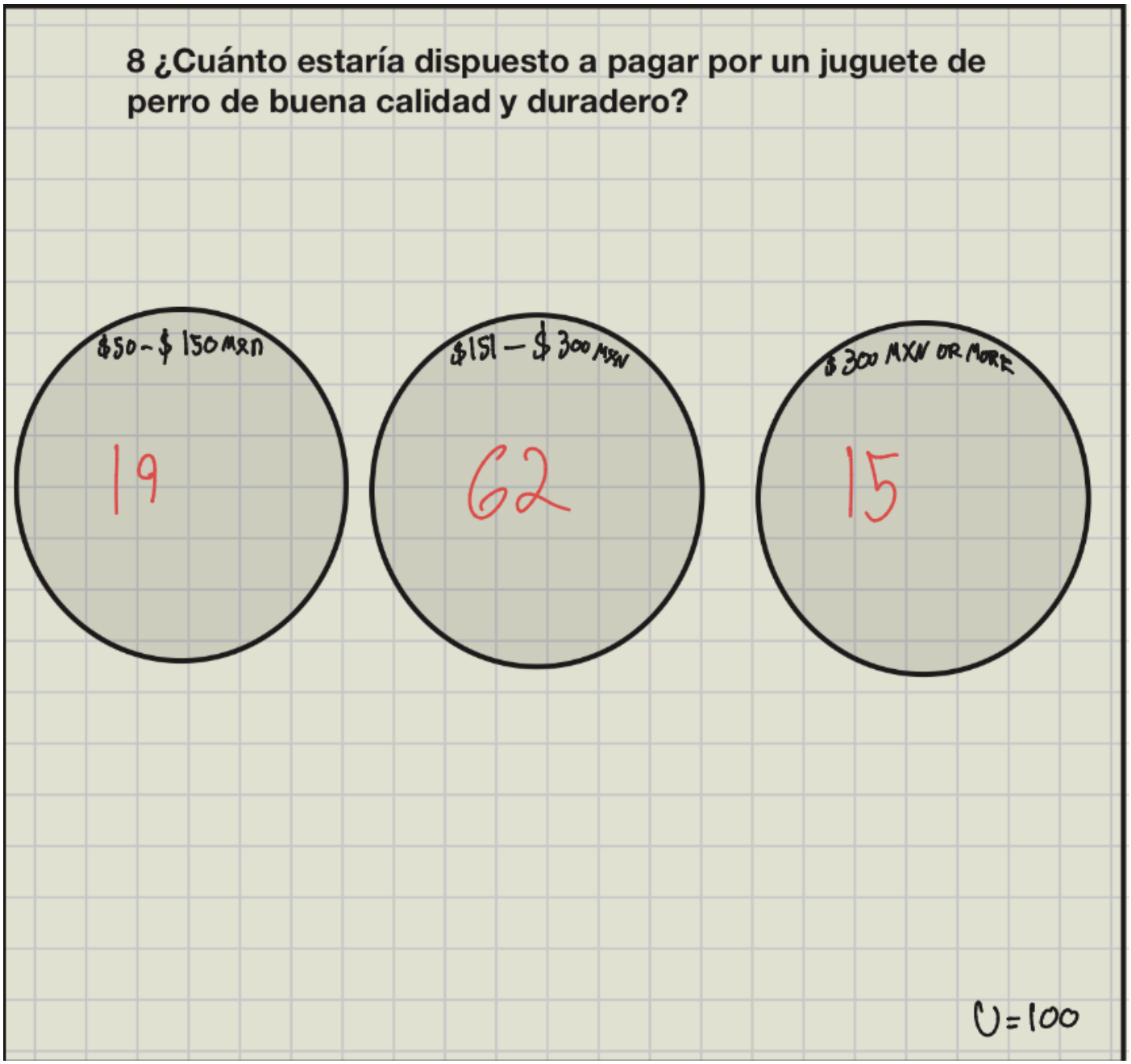
This question analyzes the main factors people consider when choosing a dog toy. The most important factors identified were durability/resistance, safe and non-toxic materials, and price. A Venn diagram was used to visualize how these factors overlap, showing that many respondents consider more than one factor at the same time when making a purchase decision.

Set Diagram Explanation

The Venn diagram represents the relationship between durability, safety, and price. Each intersection shows the number of respondents who selected more than one factor, while the central region represents those who consider all three factors simultaneously.

Logical Propositions

- If a respondent values durability, they are likely to also value safe and non-toxic materials.
- If price is important, durability is often important as well.
- The intersection of durability, safety, and price represents the core decision criteria for dog toy purchases.



Question 8 – Quantitative Analysis

This question analyzes how much respondents are willing to pay for a high-quality and durable dog toy. The results show that most respondents are willing to pay between **$151 and $300 MXN**, indicating a strong preference for mid-range priced products. Smaller groups are willing to pay either less than $150 MXN or more than $300 MXN.

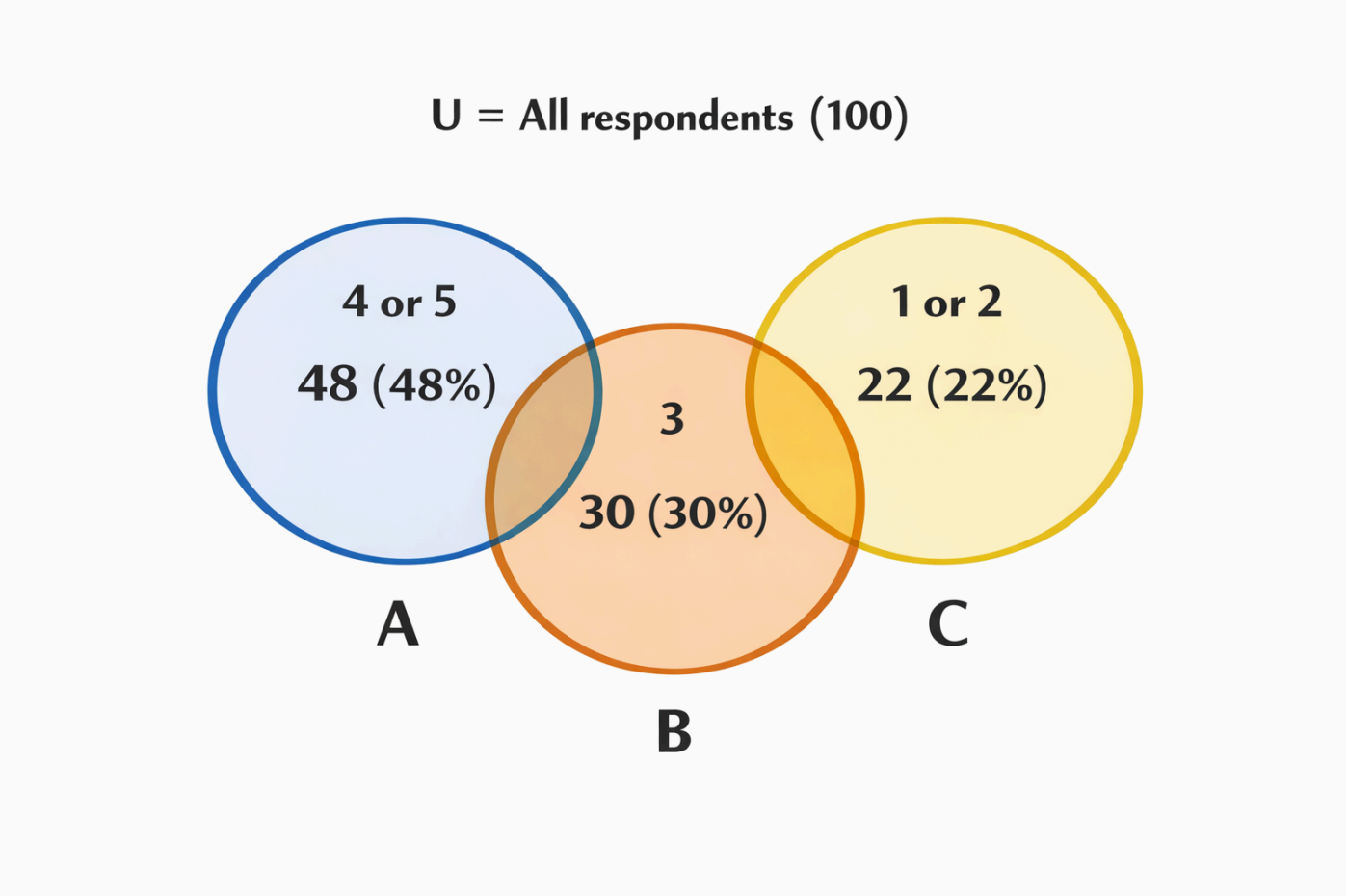
Set Diagram Explanation

The set diagram represents three independent price ranges: $50–$150 MXN, $151–$300 MXN, and more than $300 MXN. Since each respondent selected only one option, the sets do not intersect. The size of each circle reflects the number of respondents in each price category.

Logical Propositions

- If a respondent is looking for quality and durability, they are most likely willing to pay between $151 and $300 MXN.
- The majority of respondents prefer a mid-range price rather than the lowest or highest option.
- A smaller segment of respondents is willing to pay a premium price for higher-quality dog toys.

Massimo:



Importance of buying toys

Total number of people interviewed: | U | = 100

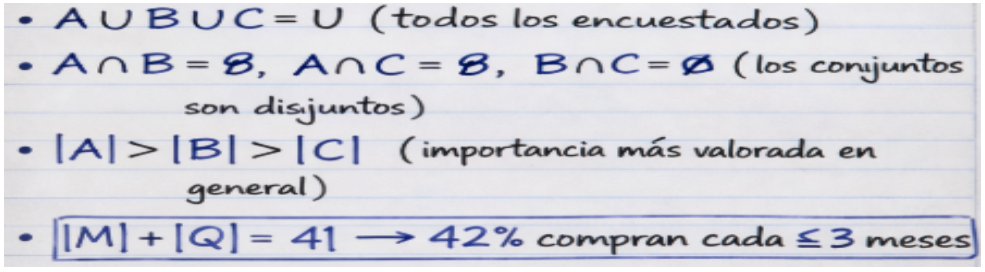
Define subset:

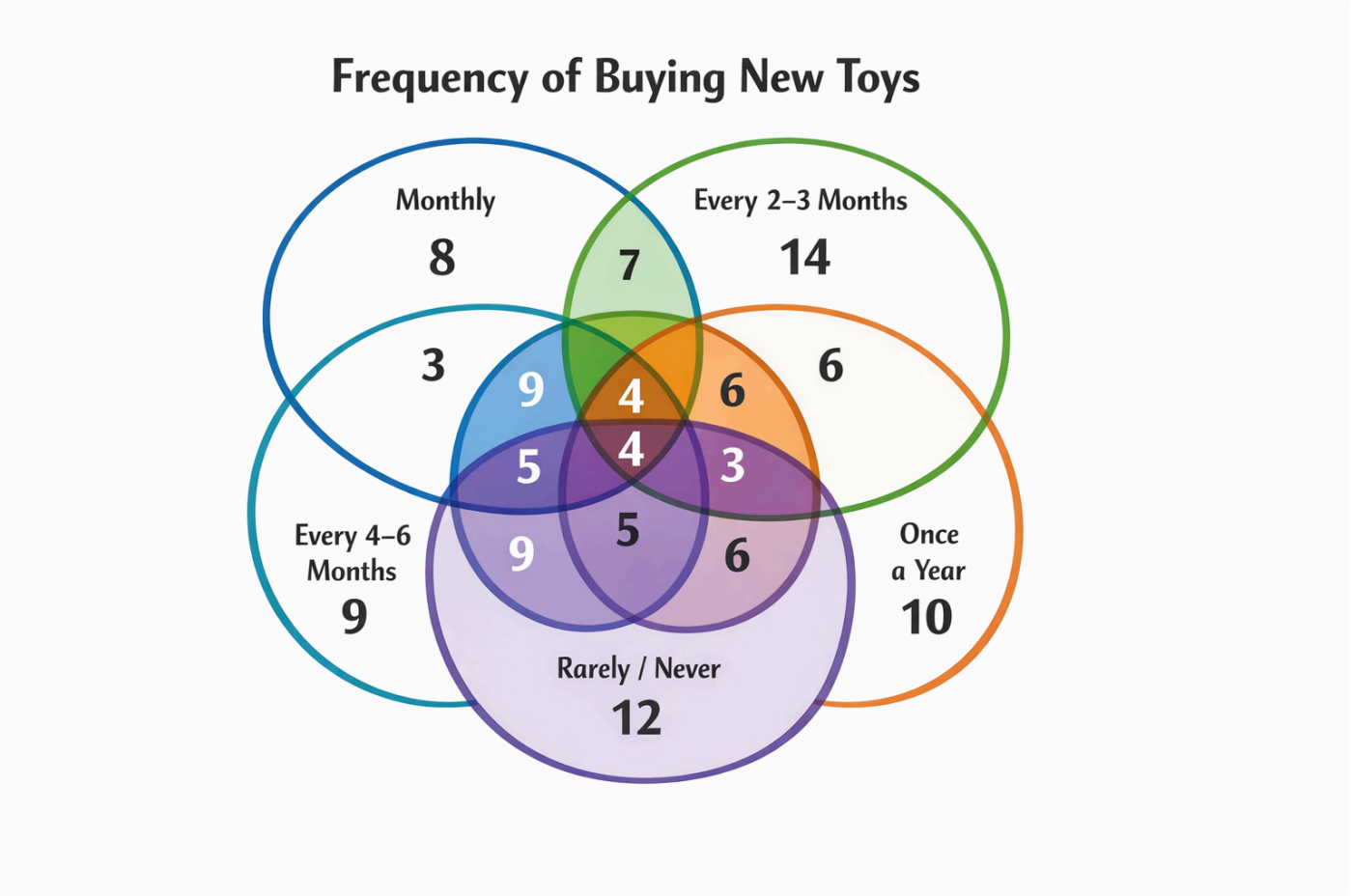
- A = { x E U | importance = 4 or 5}
- B = { x E U | importance = 3}
- C = { x E U | importance = 1 or 2}

Cardinality

| A | = 48 | B |= 30 | C| = 22

Logical Propositions:





Frequency of buying new toys

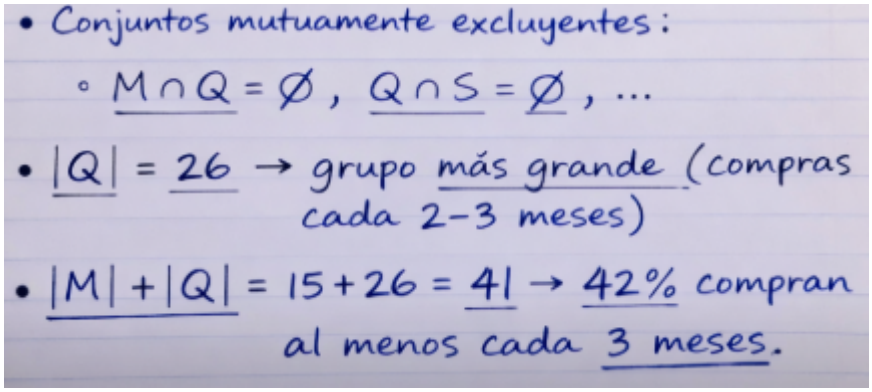
Total of interviewed people: | U | = 97

Define subsets:

- M = { x E U | buys monthly}
- Q = { x E U |{buys every 2–3 months}
- S = { x E U | buys every 4–6 months}
- Y = { x E U | buys once a year}
- R = { x E U | rarely or never buys}

Cardinality

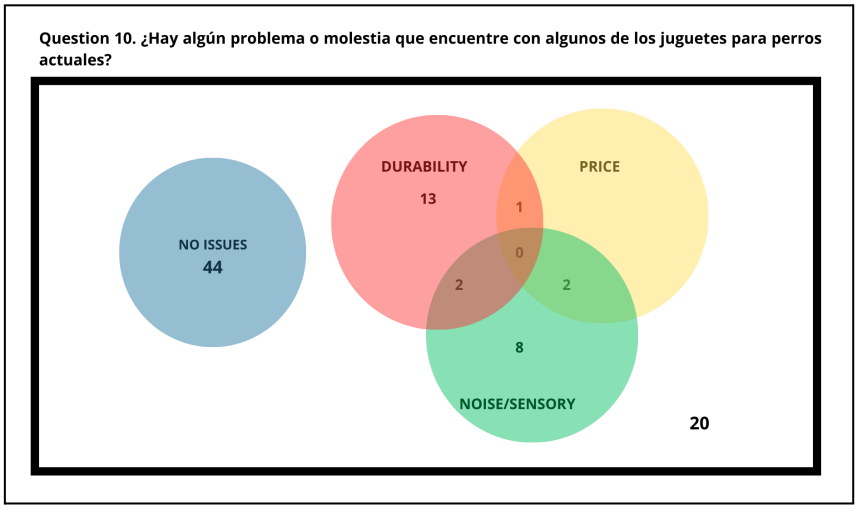
|M|=15 |Q|=25 |S|=18 |Y|=18 |R|=20



Jolett Garza:



When asked how much time their dogs spend alone, the majority of respondents an amount of time in a range of 6 to 5 hours per day. In contrast, 42 respondents left their dogs alone for 4-3 hours , while 12 only leave their dogs alone for about 1-2 hours. Only 4 respondents spend all day with their dogs.



This was an open-ended question that had a wide variety of responses. To analyze this question in an easier way, the answers were grouped into categories based on the most repeated issues. When asked whether they experienced problems with modern dog toys, most respondents indicated that they did not have any issues.

Thirteen responses referred to durability problems, such as toys breaking easily or not lasting over time. Eight responses were related to noise and sensory issues, including excessively loud sounds, materials that may be uncomfortable for dogs, or toys that produce unpleasant odors. Additionally, 20 respondents did not provide an answer useful for analyzing this question.

Intersections identified in the survey include the following:

D ∩ P = 1, indicating one response that mentioned both durability and price issues.
D ∩ N = 2, indicating two responses that mentioned both durability and noise/sensory issues.
N ∩ P = 2, represents two responses that included both noise/sensory and price issues.
N ∩ D ∩ P = 0, as no responses included durability, noise/sensory, and price issues at the same time.

7. Product Proposal

Explain in DETAIL the product you are going to implement, justifying its design based on data analysis and/or research conducted.

We present to you a product from Bark & Play, The Dog's Stress Ball, a durable soccer-style ball made of hexagonal panels with a central compartment designed to hold small treats or food. The ball includes a side opening that allows treats to be inserted and gradually released as the dog rolls and pushes it. The design of this product is directly based on the survey data collected. Out of 99 respondents, 87 reported owning dogs, making dogs the primary target market. Additionally, the age distribution shows that most dogs fall into the Adult Young and Adult categories, which are typically high-energy stages requiring both physical and mental stimulation.

The survey also analyzed what factors consumers consider most important when purchasing a dog toy. The most frequently selected factors were durability, safety, and mental stimulation. Price and design were also mentioned, but functionality and resistance were clearly prioritized.

Design Justification Based on Survey:

Durability: Since durability was one of the most important factors identified in the Excel analysis, the ball is made from resistant, non-toxic materials capable of withstanding chewing and active play. This responds directly to consumer concerns about toys breaking easily.

Safety: Safety was another key factor. The toy is made from pet-safe materials, with smooth edges and a secure internal compartment to prevent choking hazards. This ensures the product aligns with the expectations expressed in the survey.

Mental Stimulation: The treat-dispensing compartment directly addresses the demand for interactive and stimulating toys. As the dog pushes and rolls the ball, treats are released gradually, encouraging problem-solving behavior and keeping the dog engaged for longer periods.

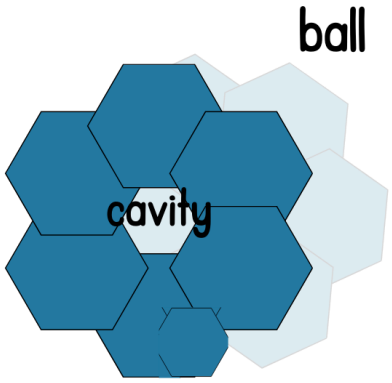
Physical Activity: The spherical soccer-style hexagon design allows unpredictable rolling movement. This stimulates chasing and active play, which is especially important considering that the majority of surveyed dogs are in energetic age ranges.

By combining durability, safety, physical activity, and brain engagement, The Dog's Stress Ball is not just designed but strategically built with the data obtained from the survey. The product directly responds to the needs and preferences identified in the Excel analysis, making it a data-driven solution rather than an idea.

7.1 Sketch of the proposed solution

Sketch your proposed solution so that you have a visual idea of what you want to achieve in this project. If you used AI tools add the link to the chat or the prompts.

Sketch #1:



Sketch #2 Designed with CHATGPT (link at appendices)

https://www.geogebra.org/classic/eeczdmwj



8. Model in Matlab

8.1 Model code

DETAILED explanation of the code, clearly indicating the person who performed each code segment.

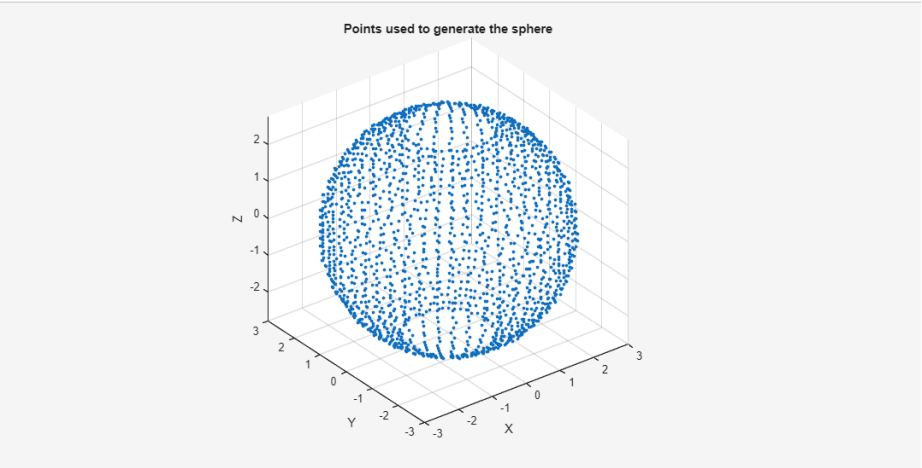
Matlab file with the code including:

- Team information
- Specific comments on the code.
- Variable descriptions
- Detailed explanation of the steps followed in the solution
- Name of the person who created EACH section of the code (INDIVIDUAL GRADE)

8.2 Matlab model vs. original proposal

Image comparing the Sketch of section 7 with the model achieved in Matlab indicating the adjustments that were made and the reasons for them.

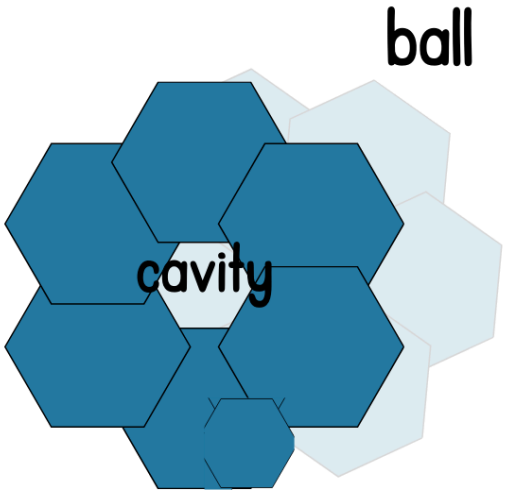
- Image indicating the points to create the graphic.

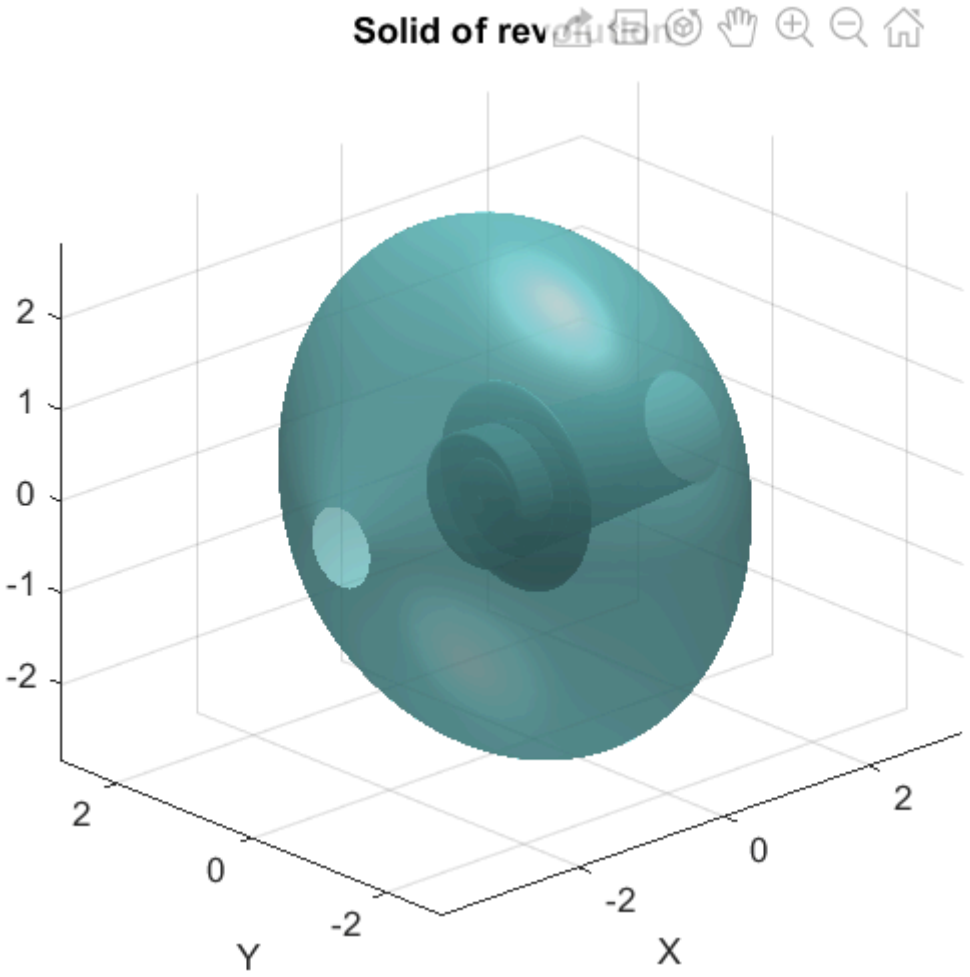



```
nPoints = 50;
[phi,theta] = meshgrid(linspace(0,pi,nPoints), linspace(0,2*pi,nPoints));

X = R * sin(phi).*cos(theta);
Y = R * sin(phi).*sin(theta);
Z = R * cos(phi);
```

- Image comparing the proposed sketch with the actual Matlab simulation.





- Description of the differences between the original model and the simulation.

Some adjustments were made when converting the sketch into a MATLAB model. The sketch was more like the general idea of a sphere with a hole and segmented surfaces, but it was not very precise. In MATLAB, the sphere was generated using parametric equations that give us symmetry and consistency, since a drawn sketch cannot show exact proportions.

Another important adjustment was how the hole was created. In the sketch, the hole was only visual, but in the simulation it was implemented by removing points that meet a radial condition. This was done to make the hole centered accurately.

The segmented appearance was also changed. While the sketch shows a segmented surface, in MATLAB this effect was done by changing coordinate values in angular sections. This approach keeps the original visual idea while making the model easier to generate in matlab.

9. Cost analysis

The material selected for the dog toy is natural rubber because it is commonly used in chew toys due to its flexibility, durability, and resistance to bites.

The volume of the object was obtained using the MATLAB model developed in this project. Once the volume was calculated using numerical integration, the mass of the toy was estimated using the density of natural rubber.

The density used in the calculation was 1.10 g/cm³. Using the calculated mass, the amount of material required to manufacture one unit of the toy was determined.

To estimate the material cost, an average market price of natural rubber in Mexico was used. The value used in this analysis was approximately 43.66 MXN per kilogram of natural rubber.

Additional production costs were also considered to obtain a more realistic manufacturing estimate. These include labor costs, machine energy consumption, finishing processes, and packaging of the final product.

Explanation of the Cost Calculations

The cost calculations were based on the physical properties obtained from the MATLAB model.

First, the profile of the dog toy was defined using outline points and polynomial interpolation. The profile was then rotated around the x-axis to generate a solid of revolution representing the three-dimensional shape of the toy.

The volume of the object was calculated numerically using the trapezoidal integration method implemented in MATLAB with the trapz function. The mathematical expression used is:

$V = \pi \int r(x)^2 \, dx$

where r(x) represents the radius of the profile.

Once the volume was obtained, the mass of the object was calculated using the density of natural rubber. The mass was converted from grams to kilograms.

The material cost was calculated by multiplying the mass of the toy by the price of natural rubber per kilogram. Finally, additional production costs including labor, energy consumption, finishing, and packaging were added to estimate the total manufacturing cost per unit.

After including all these factors, the estimated manufacturing cost per unit is approximately 60–120 MXN. Based on this production cost and considering the prices of similar dog toys in the market, the product will be sold for approximately 240 MXN per unit. This price also aligns with the survey results, where most respondents indicated they are willing to pay between 151 and 300 MXN for a durable dog toy. This allows the product to remain competitive while covering production costs and generating a reasonable profit margin.

References

Engineering Toolbox – Natural Rubber Properties

https://www.engineeringtoolbox.com

Kari. (2026, January 23). *How to choose wholesale dog items: A complete buying guide*. SmartBuy Alibaba. https://smartbuy.alibaba.com/buyingguides/wholesale-dog-toys

Selina Wamucii – Natural Rubber Price Data in Mexico

https://www.selinawamucii.com

10. Proposal for technological innovation (at least one proposal per person)

In this section, you should mention the technological innovations that each of the team members propose for the product, as well as their features. Explain fully how each of the sensors/actuators in the solution will behave. This description should contain the general idea and general interactions.

Person who proposes it	Technological innovation	Expected functionality
Sebastian Huerta	A remote control vibrator inside the ball that lets it roll and move how the owner wants	The ball will contain a small vibration motor connected to a Bluetooth receiver and a controller. Thanks to a mobile app or remote control, the owner can activate the motor to generate controlled vibrations. These vibrations will move inside the ball, allowing it to roll,

		spin, or change direction thanks to the vibration. The ball will also include a rechargeable battery and basic motion sensors that detect when the ball is moving , helping control the power usage and adjust the vibration. The interaction works like this: the user sends a command from the remote/app the controller receives the signal the vibration motor activates the internal vibration causes the ball to move in the desired direction. This creates an interactive toy that the owner can control remotely.
Massimo Brambilla	Mini speakers that lets you put whatever sound you want for your dog	The speaker will contain a Bluetooth receiver and a programmable sound controller. Through a remote interface, the owner can upload or select different sounds that best suit their dog's preferences. These sounds can range from squirrel sounds to playful effects, all customizable directly from the code. The speaker will also include a rechargeable battery. The interaction works like this: the user sends a command from the app, the controller receives the signal, the speaker activates and plays the chosen sound, and the dog responds to the audio stimulus. This creates an interactive device that the owner can control remotely, tailoring the sound experience to their pet's needs.
Sebastian Noriega	Smart dog toy with treat dispenser activated by a button	The toy will include a small compartment with dog treats. When the owner presses a button on the toy, a small mechanism opens and releases a

		treat. The system will use a small motor or servo controlled by a microcontroller and powered by a rechargeable battery. This allows the owner to reward the dog during playtime.
Andres Sarmiento	Motion sensor for activity detection	This proposal consists of integrating a small accelerometer or motion sensor inside the toy. The sensor would detect when the dog is interacting with the ball by measuring movement such as rolling, shaking, or pushing. The idea is that the system continuously monitor activity. If the toy detects that it's been inactive for a long time, it could activate a function like vibrating, or a light signal to attract the dogs attention. This would help maintain the dogs engagement, and reduce boredom when the owner is not present.

11. Project Conclusions and Insights (individual)

11.1 Project conclusions

11.2 Individual insights

- What would you do differently in terms of:
    - Survey design
    - Survey questions
    - Survey analysis
    - Product Proposal
    - Project management
    - Programming
- Did your perspective of the project change after having a physical prototype compared to just creating a computer model?

12. Functional Prototype

- Description of the integrated functional prototype
- Image of the integrated prototype
- Video explaining the operation of the integrated functional prototype

13. Project Management Document

Shared document that includes: project phases, SPECIFIC activities, estimated times and those responsible

- Link to the project management document, you can use the following template:

  https://lucid.app/lucidchart/0320bcb7-db42-4834-b06d-cc50c9b29c06/edit?viewport_loc=-1341%2C-510%2C4009%2C2420%2CAgEl0s_1ViQ_&invitationId=inv_4c183629-a09c-4f8a-b31b-b070cb66d4ba

WEEKLY PLANNER:
https://www.canva.com/design/DAHBgLiv2sc/p8QFP3yPJXRFw5gygHGxuQ/edit?utm_content=DAHBgLiv2sc&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

14. References

Please note the references you used in IEEE format (you can use this widget: https://www.citethisforme.com/citation-generator/ieee)

Another option is to use Zotero/BibDesk/Mendely or directly the references generated by electronic libraries.

Lawson, L. (2025, January 20). Why mental stimulation is crucial for dogs. Dogster. https://www.dogster.com/dog-health-care/why-mental-stimulation-is-crucial-for-dogs

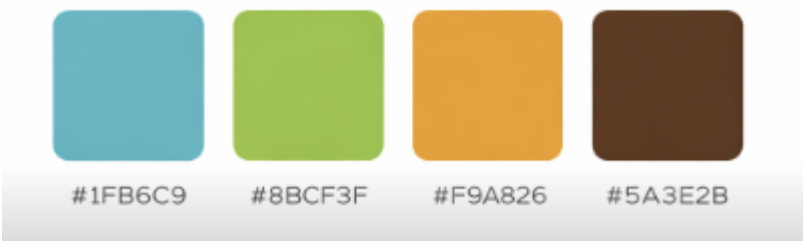
3Mitropoulos, T., & Andrukonis, A. (2025). Dog owners' job stress crosses over to their pet dogs via work-related rumination. Scientific Reports, 15, Article 1131. https://www.nature.com/articles/s41598-025-01131-x

Poian, A. (2026, January 16). Dog enrichment score calculator. Dog City Guide. https://dogcityguide.com/articles/dog-enrichment-score-calculator-2026/

Valladares, R. (2024, July 24). Perros, gatos o cuyes suelen ser parte de las familias como un integrante más [Blog post]. Milenio Finanzas. https://mileniofinanzas.wordpress.com/2024/07/24/perros-gatos-o-cuyos-suelen-ser-parte-de-las-familias-como-un-integrantes-mas-no-solo-como-mascotas-en-inmuebles24-hicimos-una-encuesta-entre-nuestros-usuarios-para-conocer-mas-sobre-como-influye

15. Appendices

Color palet:



Sebastian huerta 11 Feb conversation with chatgpt:

https://chatgpt.com/share/698cb2aa-dcbc-8000-8c5d-2fb24186338f

Massimo conversation with chatgpt:

https://chatgpt.com/share/699a4f8e-b424-8008-9d6d-6896b42738a0